# Explainable AI Final Assignment

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# 1 Introduction

With the introduction of machine learning models into more higher stake domains such as healthcare the demand has risen for transparency and interpret ability. While complex models might have good prediction scores these mostly operate as so called "Black boxes". This therefore prevents transparency, which can cause problems with trust and ethical deployment. Therefore this project aims to improve the interpretability of the Pima Indians Diabetes dataset by applying two Explainable Artificial intelligence methods. The first one being SHAP (SHapley Additive exPlanations) which is a feature attribution method and the second one being DiCE (Diverse Counterfactual Explanations) which is a method to provide counter factual explanations. This paper aims to unify these methods to see if together they can provide a better explanation of the primary risk factors in the Pima Indians diabetes dataset. This will be done through the use of a practical implementation in the form of a python streamlit dashboard which serves as a front end for a backend built upon python packages.

## 2 Research Question

This will condense down into the research question:

> "To what extent do SHAP attributions and DiCE counterfactuals provide a unified explanation of the primary risk factors within the PIMA Indians diabetes dataset?"

# 3 Methodology

## 3.1 SHAP

SHapley additive exPlanations (SHAP) are a feature-based interpretability method which can be integrated into supervised ML models to gain better understanding of predictions, enhancing the transparency and the trustworthiness of models(Ponce-Bobadilla et al., 2024) The mathematical formula was found by Lloyd Shapley and can be deduced from the following

$$\phi_i(f, x) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)] \tag{1}$$

where $N$ represents the complete set of input features, $S$ denotes a subset of features that does not feature $i$ and $f(s)$ is the expected value of the model prediction conditioned on the feature subset(Shapley et al., 1953).

## 3.2 DiCE

Diverse Counterfactual Explanations (DiCE) was developed by Mothilal et al. (2020) in collaboration with Microsoft. The goal of the DiCE framework is to create a Counterfactual Explanations Framework which can be used to within Python to show counterfactual explanations. The idea behind this is that counterfactual implementations which are a type of Post-Hoc explanations of Machine Learning Model can be used as a hypothetical example to show how different predictions can be obtained. Counterfactual explanations are defined by Wachter et al. (2017) as showing feature-preturbed versions of the same person who would have received a loan when for example: "your income was higher by $10,000". However within the original DiCE python framework some limitations were found with regards to balancing proximity, diversity and model robustness. This is especially concerning when using Counterfactual explanations (CF) in high stakes decision making contexts such as healthcare, finance and education. Where there is a high need for precise, interpretable, and adaptable CFs (Laugel et al., 2023).

The mathematic notation for the DiCE optimization formula is:

$$\begin{aligned} C(\boldsymbol{x}) = \underset{\boldsymbol{c}_1,\ldots,\boldsymbol{c}_k}{\arg\min} \frac{1}{k} \sum_{i=1}^{k} \text{yloss}(f(\boldsymbol{c}_i), y) + \frac{\lambda_1}{k} \sum_{i=1}^{k} dist(\boldsymbol{c}_i, \boldsymbol{x}) \\ - \lambda_2 \, \text{dpp_diversity}(\boldsymbol{c}_1, \ldots, \boldsymbol{c}_k) \end{aligned} \tag{2}$$

This formula is structured as the following: $c_i$ is a counterfactual example (CD), k is the total number of CFs which need to be generated. $f(.)$ is the ML model, $yloss(.)$ is a metric that minimizes the distance between $f(.)'s$ prediction for $c_i$ and the desired outcome of $y$, $d$ is the total number of input features, x is the original input, and dpp_diversity(.) is the diversity metric $\lambda_1$ and $\lambda_2$ are hyperparameters that balance the three parts of the lossfunction.

## 3.3 Code Logic

The code logic is build up by the following logic:

1. Initialization of the data
2. Create train test split
3. Scale the data
4. Train models
5. Create SHAP and DiCE explanations
6. Create predictions
7. Create visuals for SHAP
8. Initialize the DiCE Counterfactual explanations
9. Create table for DiCE and Visual

## 3.4 Tech Stack

Within this project a Python implementation was chosen due to the good integration packages. The packages which were used can be distinguished into 3 different categories. Explainable AI libraries being SHAP and diceml. Shap being a model which uses Logistic Regression from SKlearn and calculates the feature attributions which can be used. Whilst diceml is used to create Counterfactual Explanations, meaning that dice provides counterfactuals to show how changing input features could alter the outcomes. The second being the sklearn package which is used for Modeling through Logistic Regression and processing the data through StandardScaler and train test split. For the data manipulation the pandas and numpy packages were used. For the frontend streamlit was used to create a frontend dashboard which provides an interactive element whilst plotly and matplotlib create visual graphs. The reasoning behind this techstack was that through these packages both the novel and the lecture methods could be explained.

## 3.5 Dataset

The dataset Pima Indians Dataset was used to create this dashboard. This dataset is licensed under the CC0 dataset by the UCI Machine Learning on Kaggle. The goal of this daatset is to diagnose whether a patient has diabetes based on certain measurements. All these measurements were taken from female patients which were at least 21 years old and of Pima Indian heritage.

### 3.6 Comparison of SHAP and DiCE

It is important to remember when comparing SHAP and DiCE that these methods serve two different purposes. SHAP serves as a feature attribution method whilst DiCE serves to find counterfactual explanations. This difference must be taken into account when comparing faithfulness, stability and completeness. The faithfulness of SHAP is high due to its fair distribution among its features whilst DiCE can be variable due to it being prone to sometimes recommend implausible changes such as "changing age". The stability of SHAP is higher than in DiCE due to its reliance on mathematics whilst DiCE can react significantly to small changes in input. The completeness of SHAP is also higher than DiCE due to it not always explaining the entire prediction but rather the minimum changes needed to create the counterfactual explanation. To conclude we can see that SHAP suffers from the correlation trap meaning that if values are correlated SHAP can fail to capture this relationship whilst DiCE suffers from a plausibility gap meaning that if for example the counterfactual explanations which are provided are not plausible, this could make the values less trustworthy.

# 4 Results

We will discuss one entry from the dataset which is patient index 18 on the slider. This patient patient has a probability of 78% for diabetes. This will show a case of a unified explanation. The high SHAP values of the feature Glucose show that this is the most important contributor to the risk. When we look at the counterfactual explanation we can see that the value for Insulin shoots up meaning that to change this scenario the value for insulin should increase the fastest. Therefore indicating that to flip this patient to low risk the value for insulin should increase. This therefore suits the real life human intuition that if people have too high blood sugar they might have diabetes. If this is the case it is necessary to increase the level of insulin in the blood to decrease the blood sugar values. Thus we find that this suits the human intuition.

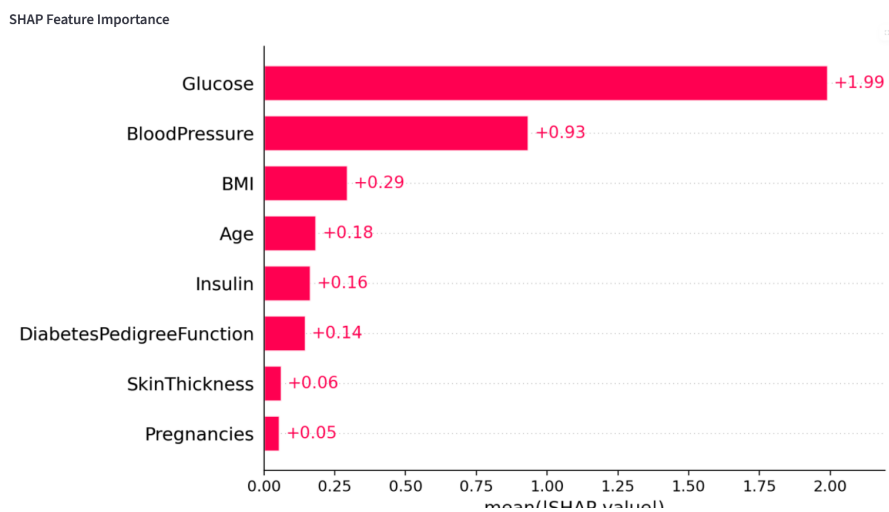


Figure 1: SHAP feature importance

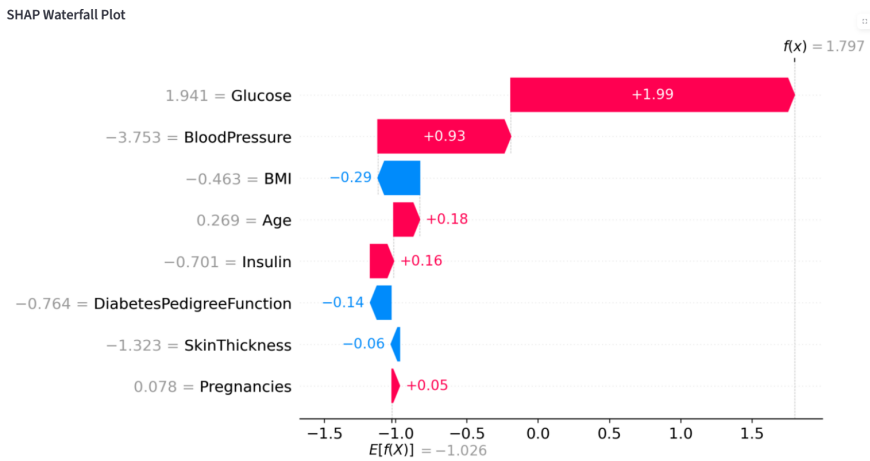


Figure 2: SHAP Waterfall plot

Counterfactual Explanations Table

| | Pregnancies | Glucose | BloodPressure | SkinThickness | Insulin | BMI | DiabetesPedigreeFunction | Age | Outcome |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 0.0777 | -3.2006 | -3.7527 | -1.3228 | 6.4049 | -0.4633 | -0.7641 | 0.2691 | 0 |
| 1 | 0.0777 | 1.9415 | -0.6433 | -1.3228 | 5.3129 | -0.4633 | -0.7641 | 0.2691 | 0 |
| 2 | 0.0777 | -1.5354 | -3.7527 | -0.2275 | -0.7012 | -0.4633 | -0.7641 | 0.2691 | 0 |



Figure 3: Counterfactual explanations

# 5 Discussion

We find if we look at the results that through the combination of SHAP and DiCE do provide a unified explanation of the primary risk factors within PIMA Indians diabetes dataset. This is due to the fact that the most important features for diagnosing diabetes are shown in the. This conclusion can therefore be transferred to the PIMA Indians diabetes dataset which is used where patients who are likely to have diabetes can easily see which factors are most important for their diagnosis of diabetes and what can be done to improve the condition.

The main pitfalls for this project would be that both SHAP and DiCE have some limitations. SHAP suffers from the correlation trap an example from this dataset is that BMI, Skinthickness and Insulin are all biologically related however SHAP can struggle with this correlation as it rather splits the importance between these values which can mislead the doctor which works with this. The main pitfall of DiCE is that the counterfactual explanations can suggest changes which are not plausible for example "decreasing age" which is not possible.

Within this research the main problem with bias is that the dataset only features females within the PIMA Indians diabetes dataset. This therefore proposes the problem that the population is biased. The second problem with this dataset is that the population within the dataset only consists of people with a PIMA Indians background. This creates bias as this group could be more prone to have diabetes.

There are some ethical considerations. The first one being the dataset which only has PIMA Indians which shows that this was the only group which was considered. This does lead to a specific target group but questions can be asked on whether this data was collected ethically.

A recommendation which can be made for future research is that the DiCE is an older framework with some structural limitations such regarding the balance of proximity, diversity and model robustness. A solution to these limitations can be found in DiCE extended (DiCE-X) which is an extended CF framework which alters the Optimization function to balance proximity, diversity and feasibility of counterfactuals (Bakir et al., 2025). For future research it would be recommended to use DiCE-X with further optimization of the values of the optimization function to further improve results.

# 6 Conclusion

To conclude it is important to mention that the combination of SHAP values and DiCE can be used to offer an explanation about the primary risk factors within the PIMA Indians diabetes dataset. This is due to that fact that these values do provide a unified explanation regarding the hierarchy of the risk factors in the dataset but that there can be some differences in the practical outcomes. However it must be noted that this methodology still possesses some major structural flaws with regards to bias, pitfalls and ethical considerations. These flaws must be taken into account when considering these methods and in future research.

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